Unified ACT: shared observations, two task interfaces

One model family across garment folding, bowling and hook pull · model schema unified-act-v1

(a) Shared causal observation backbone

RGB observations
Current I_0: full grid
Past $I_{-H:-1}$: pooled
Same causal timestamps

ResNet-18
Shared per camera
Current: 15 × 20*
Past: 4 × 4 / frame

Visual tokens
Native spatial PE
Physical-time PE
History / camera masks

ACT encoder
4 layers · d = 512
8 heads · FFN 3200
Native core retained

calibrated bilinear sample

K observed tracks
Current anchor $x_i(0)$
Past displacement Δx_i
Time + validity

Point encoder
Shared MLP: 128 → 128 → 128 → 512 (H = 16)
Aligned image features

Robot state
Observed q_0 only
Train-only stats

Gated point fusion
Q: ACT memory
K,V: point tokens
M′ = M + tanh(g) Attn

Point encoder is permutation-equivariant over identities.
Stationary control keeps anchors, time, masks and architecture;
only past displacement is set to zero, before image sampling.

* 480 × 640 input. Resolution, joints, actions and timing are task adapters.
No 3NN prior, future GT input, object ID or rigid post-processing.

M′(z)

M′(0)

(b) Policy interface
Native ACT decoder · 1 layer
100 learned action queries*
Native linear action head

Action chunk
$\hat{a}_{0:99} \in R^{100xA}$
Lpolicy = masked L1 + 10 KL
Validate final raw action MSE

(c) Dynamics interface
XYZ + future-time queries
Independent point cross-attn
Given action tokens in head

Future point motion
$\Delta\hat{x}_i(t)$, t > 0
Primary: point SmoothL1
Evaluate EPE / moving EPE

Committed controls
u_0:T + physical time
Dynamics head only

M′(0)

(d) Optional completion
XYZ + past-time queries
Separate point cross-attn
Observed anchors; hidden past

Past point motion
$\Delta\hat{x}_j(t)$, t < 0
t = 0 excluded from target
Default: auxiliary OFF

ACT parity

Single-frame variant bypasses extensions.
Same native weights, objective and action head.

Posterior isolation

Policy: $z \sim q(z \mid q_0, a^*)$ in training; z = 0 at test.
Point heads re-encode observations with z = 0.

Auxiliary gradient control

Detached probe first; bounded sharing is optional.
Full-update shared aux / primary norm ≤ 0.1.

Architecture implementation only · no new trained-model performance is claimed · * task-specific horizon